

Ejercicios

Sección 11.2 Centro de gravedad

11.1) $m_1 = 0.120 \text{ kg}$ $m_2 = 0.055 \text{ kg}$ $m_3 = 0.110$

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2 + m_3 x_3}{m_1 + m_2 + m_3} = \frac{(0.120)(0.250) + 0 + (0.110)(0.500)}{0.120 + 0.055 + 0.110} = 0.298 \text{ m}$$

29.8 cm a la derecha del extremo izquierdo.

11.3) $x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$

$$1.2 = \frac{(7.8)(1) + (2.4)x_2}{7.8 + 2.4}$$

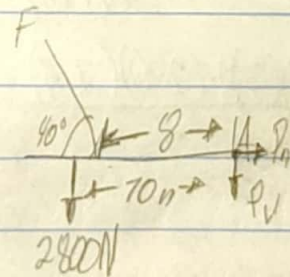
$$x_2 = \frac{(7.2)(7.8 + 2.4) - (7.8)(1)}{2.4} = 1.35 \text{ m}$$

Sección 11.3 Solución de problemas de equilibrio de cuerpos rígidos

11.5) Levantamiento de una escalera

$$F(8\text{ m})\sin 40^\circ = (2800)(10)$$

$$F = 5.45 \text{ kN}$$



11.7) el peso motor = 1000 N, por la suma de fuerzas

$$\frac{(2\text{ m})(600\text{ N})}{(1000\text{ N})} = 1.2 \text{ m}$$

$$b) \text{ peso motor} = 400\text{ N} + 600\text{ N} - 200\text{ N} = 800\text{ N}$$

$$(600\text{ N})(2\text{ m}) = (200\text{ N})(1\text{ m}) + (800\text{ N})x$$

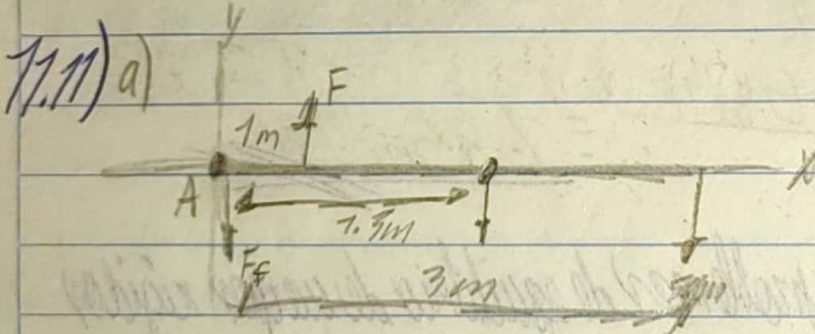
$$x = 1.25\text{ m} \quad \dots \text{centro de gravedad en } 0.75\text{ m}$$

$$7.9) a) \sum F_y = 0 \rightarrow T_A + T_B - W - W_{\text{barra}} = 0$$

$$W = T_A + T_B - W_{\text{barra}} \quad 500\text{ N} + 400\text{ N} - 350\text{ N} = 550\text{ N}$$

$$b) \sum \tau = 0 \quad T_B(1.5\text{ m}) - Wx - W_{\text{barra}}(0.75\text{ m}) = 0$$

$$x = \frac{T_B(1.5\text{ m}) - W_{\text{barra}}(0.75\text{ m})}{W} = \frac{(400)(1.5) - (350)(0.75)}{550} = 0.619\text{ m}$$



$$\sum \tau_A = 0 \quad F(1\text{ m}) - (500\text{ N})(1.5\text{ m}) - (280\text{ N})(3\text{ m}) = 0$$

$$F = \frac{(500)(1.5) + (280)(3)}{1} = 7420\text{ N}$$

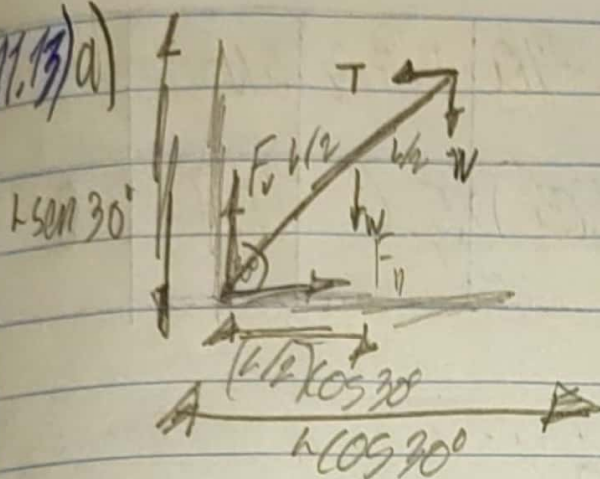
$$b) \sum F_y = 0$$

$$F - F - 280\text{ N} - 500\text{ N} = 0$$

$$F_F = F - 280\text{ N} - 500\text{ N} =$$

$$7420\text{ N} - 280\text{ N} - 500\text{ N} = 7140\text{ N}$$

11.13) d)



$$\sum F_y = ma_y$$

$$F_v - W - W = 0$$

$$F_v = 2W$$

$$\tau_1 = 0$$

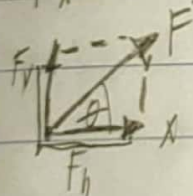
$$T \sin 30^\circ - W \left(\frac{L}{2}\right) \cos 30^\circ - W (L \cos 30^\circ) = 0$$

$$T \sin 30^\circ - (3W/2) \cos 30^\circ = 0$$

$$T = \frac{3W \cos 30^\circ}{2 \sin 30^\circ} = 2.6W$$

$$\sum F_x = ma_x$$

$$T - F_h = 0 \quad F_h = 2.6W$$



$$F = \sqrt{F_h^2 + F_v^2} = \sqrt{2.6^2 + 2^2} = 3.28W$$

$$\tan \theta = \frac{F_v}{F_h} = \frac{2W}{2.6W} = \theta = 37.6^\circ$$

b) $\sum \tau_1 = 0 \Rightarrow (T \cos 30^\circ)(L \sin 45^\circ)(L \cos 45^\circ) - W \left(\frac{L}{2}\right) \cos 45^\circ - W (L \cos 45^\circ) = 0$

$$\sin 45^\circ = \cos 45^\circ$$

$$T(\cos 30^\circ - \sin 30^\circ) = \frac{3W}{2} \quad T = \frac{3W}{2(\cos 30^\circ - \sin 30^\circ)} = 4.1W$$

$$\sum F_x = ma_x$$

$$F_h - T \cos 30^\circ = 0$$

$$F_h = T \cos 30^\circ = (4.1)(\cos 30^\circ) = 3.55W$$

$$\sum F_y = ma_y$$

$$F_v - W - W - T \sin 30^\circ = 0$$

$$F_v = 2W + (4.1W) \sin 30^\circ = 4.05W$$

$$F = \sqrt{F_h^2 + F_v^2}$$

$$F = \sqrt{3.55^2 + 4.05^2}$$

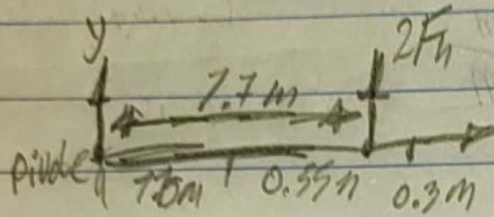
$$F = 5.39W$$

$$\tan \theta = \frac{F_v}{F_h} = \frac{4.05}{3.55}$$

$$\theta = 48.8^\circ$$

Norma

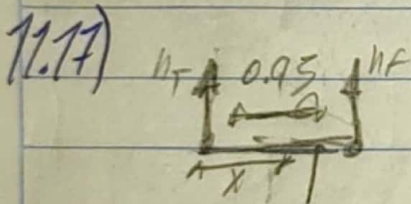
11.15) Sagartija



$$\sum \tau_z = 0 \quad (2F_h)(1.7) - W(1.5) = 0$$

$$F_h = W \frac{1.5}{2(1.7)} = 0.338 W = 272 \text{ N} \quad \sum F_y = 0$$

$$2F_f + 2F_h - W = 0 \quad F_f = \frac{1}{2}W - F_h = 402 \text{ N} - 272 \text{ N} = 130 \text{ N}$$



$$n_r = 89 \text{ N} \quad n_f = 157$$

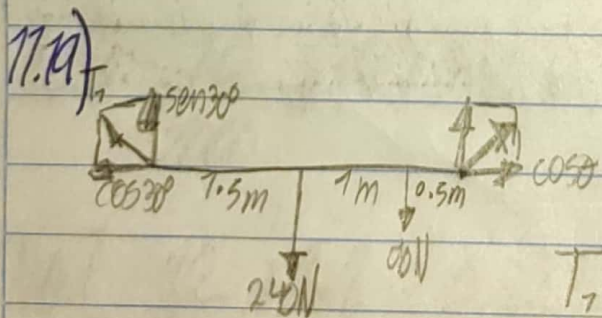
$$n_r + n_f = W$$

$$W = 246 \text{ N}$$

$$\sum \tau_z = 0$$

$$n_f(0.95\text{m}) - XW = 0$$

$$X = 0.600 \rightarrow 0.34 \text{ m} \text{ desde el VZ por delante}$$



$$\sum \tau_z = 0$$

$$(240 \text{ N})(1.5 \text{ m}) + (90 \text{ N})(0.5 \text{ m}) - (T_1 \sin 30^\circ)(1.5 \text{ m}) = 0$$

$$T_1 = \frac{360 \text{ Nm} + 45 \text{ Nm}}{1.5 \text{ m}} = 270 \text{ N}$$

$$\sum F_x = m a_x$$

$$T_2 \cos \theta - T_1 \cos 30^\circ = 0$$

$$T_2 = \cos \theta = 234 \text{ N}$$

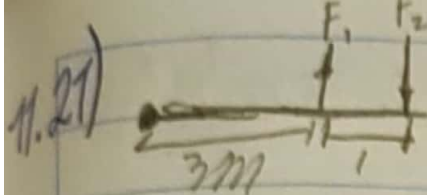
$$\sum F_y = m a_y$$

$$T_1 \sin 30^\circ + T_2 \sin \theta - 240 \text{ N} - 90 \text{ N} = 0$$

$$T_2 \sin \theta = 330 \text{ N} - (270 \text{ N}) \sin 30^\circ = 195 \text{ N}$$

$$T_2 = \frac{195 \text{ N}}{\sin 48^\circ} = 253 \text{ N}$$

Norma



$$t_1 = F_1 L_1 + (8N)(3m) = 24N \cdot m$$

$$t_2 = -F_2 L_2 = -(8N)(L+3m) = -24N \cdot m - (8N)L$$

$$\sum t = t_1 + t_2 = (24N \cdot m) - 24N \cdot m - (8N)L = -(8N)L$$

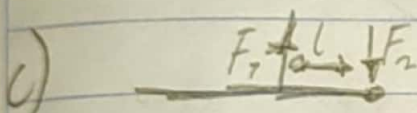
$$\sum t = -6.4N \cdot m$$

$$-(8N)L = -6.4N \cdot m$$

$$L = \frac{6.4N \cdot m}{8}$$

$$L = 0.800m$$

b) $t_2 > t_1$ = porque F_2 tiene un momento más grande



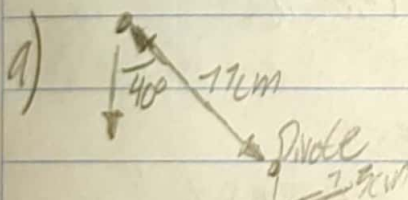
$$t_1 = F_1 L_1 = -(8N)L$$

$$t_2 = 0$$

$$\sum t = -6.4 = -(8N)L = 6.4N \cdot m$$

$$L = 0.800m$$

11.23) Músculo del cuello



b) $\sum t = 0$

$$T = \frac{W(11cm)(\sin 40^\circ) - T(1.5cm) = 0}{7.5} = 208N$$

Sección 11.4 Esfuerzo, deformación y módulo de elasticidad

11.25)

$$y = \frac{L_0 F_\perp}{A \Delta L}$$

$$A = \frac{L_0 F_\perp}{y \Delta L}$$

$$A = \frac{(2m)(400N)}{(2 \times 10^{-3})(0.25 \times 10^{-2})} = 7.6 \times 10^{-6} m^2$$

$$A = \pi r^2 \rightarrow r = \sqrt{\frac{A}{\pi}} = \sqrt{\frac{7.6 \times 10^{-6}}{\pi}} = 7.1 \times 10^{-4} m$$

$$d = 2r = 7.4 \times 10^{-3} = 7.4mm$$

$$71.27) \quad Y = \frac{10F_1}{A\Delta L} \quad A = 0.5 \text{ cm}^2 = 0.5 \times 10^{-4} \text{ m}^2$$

$$Y = \frac{(4 \text{ m})(5000 \text{ N})}{(0.5 \times 10^{-4})(0.2 \times 10^{-2})} = \underline{2 \times 10^{11} \text{ Pa}}$$

$$71.29) a) \quad Y = \frac{\text{tensión}}{\text{deformación}} = \Delta L = \frac{\tan}{Y} = \frac{F_1}{AY}$$

$$\frac{F_1}{AY} = \frac{157 \text{ N}}{(2.5 \times 10^{-2})(2 \times 10^{-1})} = \underline{3.1 \times 10^{-3}}$$

$$\frac{F_2}{AY} = \frac{98 \text{ N}}{(2.5 \times 10^{-2})(2 \times 10^{-1})} = \underline{2 \times 10^{-3}}$$

$$b) \quad \epsilon = \frac{\Delta L}{L_0} \quad \Delta L = \epsilon L_0$$

$$\text{superior} = \Delta L = (0.5)(3.1 \times 10^{-3}) = 1.6 \times 10^{-3} \text{ m} = 1.6 \text{ mm}$$

$$\text{inferior} = \Delta L = (0.3)(2 \times 10^{-3}) = 7 \times 10^{-4} \text{ m} = 1 \text{ mm}$$

$$71.33) \quad S = \frac{F_{11}}{A\theta} \quad \theta = \frac{1}{h} \quad F_{11} = F \sin 12^\circ$$

$$F = 8mg \quad m = 10 \text{ Kg} \quad 1 \text{ rad} = 180^\circ$$

$$\theta = \frac{F_{11}}{AS} = \frac{8mg \sin 12^\circ}{(10 \times 10^{-4})(12 \times 10^9 \text{ Pa})} = 0.149 \text{ rad} = \underline{8.6^\circ}$$

$$71.31) \quad \frac{\Delta V}{V_0} = \frac{\Delta P}{B}$$

$$a) \quad \Delta P = -B \frac{\Delta V}{V_0} = -(15 \times 10^9 \text{ Pa})(-0.001) = 1.5 \times 10^7 \text{ Pa} = \underline{150 \text{ atm}}$$

b) La profundidad para un aumento de presión es de 1.5 km

Norma

$$11.35) B = -\frac{\Delta P}{\Delta V/V_0}$$

$$B = -\frac{(3.6 \times 10^6)(600)}{(1-0.45)} = 4.8 \times 10^9 \text{ Pa}$$

$$\kappa = \frac{1}{B} = \frac{1}{4.8 \times 10^9} = 2.1 \times 10^{-10} \text{ Pa}$$

11.37) a) Las fuerzas horizontales se equilibran entonces el momento ejerce una fuerza en dirección opuesta

$$A) F = \frac{A \times s}{h} = \frac{(0.600)^2 (0.25 \times 10^{-3}) (4.4 \times 10^{10})}{0.0600} = 6.6 \times 10^5 \text{ N}$$

$$C) \kappa = \frac{Fh}{As} = \frac{(6.6 \times 10^5)(0.0600)}{(0.600)^2 (0.25 \times 10^{-3})} = 1.8 \text{ mm}$$

$$11.39) \text{tension} = \frac{F_{\perp}}{A} = \frac{F_{\perp}}{\pi r^2}$$

$$= \frac{90.8 \text{ N}}{\pi (0.92 \times 10^{-3})^2} = 3.41 \times 10^7 \text{ Pa}$$

$$11.41) \frac{F_{\perp}}{A} = \frac{1}{3} (24 \times 10^8) = 0.8 \times 10^8$$

$$F_{\perp} = A(0.8 \times 10^8) = (3 \times 10^{-4})(0.8 \times 10^8) = 2.4 \times 10^4$$

$$\sum F_y = may \quad F_{\perp} - mg = \quad a = \frac{F_{\perp}}{m} - g$$

$$a = \frac{2.4 \times 10^4 \text{ N}}{7200 \text{ kg}} - 9.8 = 10.2 \text{ m/s}^2$$

Ejercicios

Sección 12.1 Densidad

$$12.1) V = (\pi R^2) L = \pi (0.01425)^2 (0.858) = 5.474 \times 10^{-4} \text{ m}^3$$

$$m = \rho V = (7.8 \times 10^3) (5.474 \times 10^{-4}) = 4.27 \text{ kg}$$

$$W = mg = (4.27)(9.8) = 41.8 \text{ N, no se necesita un resorte}$$

$$12.3) V = (5 \times 10^{-3} \text{ m})(15 \times 10^{-3} \text{ m})(30 \times 10^{-3} \text{ m}) = 2.25 \times 10^{-6} \text{ m}^3$$

$$\rho = \frac{m}{V} = \frac{0.0158 \text{ kg}}{2.25 \times 10^{-6} \text{ m}^3} = 7.02 \times 10^3 \text{ kg/m}^3 \text{ no es oro puro.}$$

$$12.5) V = \frac{4}{3} \pi r^3 \quad \text{plomo} = 11.3 \times 10^3 \text{ kg/m}^3 \quad \text{aluminio} = 2.7 \times 10^3 \text{ kg/m}^3$$

$$m = \rho V = \frac{4}{3} \pi r^3 \rho \quad r_a^3 \rho_a = r_i^3 \rho_i \left(\frac{\rho_a}{\rho_i} \right)^{1/3} = \left(\frac{11.3 \times 10^3}{2.7 \times 10^3} \right)^{1/3} = 1.6$$

$$12.7) V = \pi (R_2^2 - R_1^2) L \quad R_1 = 1.25 \times 10^{-2} \text{ m} \quad R_2 = 1.75 \times 10^{-2} \text{ m}$$

$$V = \pi ([0.0175 \text{ m}]^2 - [0.0125 \text{ m}]^2) (1.5 \text{ m}) = 7.07 \times 10^{-4} \text{ m}^3$$

$$m = \rho V = (8.9 \times 10^3) (7.07 \times 10^{-4}) = 6.29 \text{ kg}$$

$$W = mg = 67.6 \text{ N}$$

Sección 12.2 Presión en un fluido

12.9) Cálculo en Mercurio

$$a) P - P_0 = (\rho)(g)(h) = (13.6 \times 10^3) (9.8) (500) = 6.72 \times 10^6 \text{ Pa}$$

$$b) h = \frac{P - P_0}{\rho g} = \frac{7.81 \times 10^6 \text{ Pa}}{(1.03 \times 10^3) (9.8)} = 781 \text{ m}$$

Norma

$$12.11) \quad pgh = 5980 \text{ Pa}$$

$$h = \frac{5980 \text{ Pa}}{\rho h} = \frac{5980}{(1020)(9.8)} = \underline{0.587 \text{ m}}$$

12.13) Parado de abeyo

$$a) \quad pgh = (1060)(9.8)(1.85) = \underline{1.92 \times 10^4 \text{ Pa}}$$

$$12.15) \quad A = \pi r^2 \quad \Delta F = (\Delta P)A = pgh$$

$$r = 4.1 \text{ mm}$$

$$h = \frac{\Delta F}{\rho g A} = \frac{7.5 \text{ N}}{(1.03 \times 10^3)(9.8)\pi(4.1 \times 10^{-3})^2} = \underline{2.8 \text{ m}}$$

$$12.17) \quad P = P_0 + pgh$$

$$P - P_{\text{aire}} = pgh = (1 \times 10^3 \text{ kg/m}^3)(9.8 \text{ m/s}^2)(6.1 \text{ m}) = \underline{6 \times 10^4 \text{ Pa}}$$

$$12.19) \quad P = P_0 + \rho gh \quad F = \rho A$$

$$F = (\rho gh)A - W = (1.03 \times 10^3 \text{ kg/m}^3)(9.8 \text{ m/s}^2)(30 \text{ m})(0.75 \text{ m}^2) - 300 \text{ N} =$$

$$F = \underline{2.27 \times 10^3 \text{ N}}$$

$$12.21) \quad A = \pi r^2 = \pi(0.150 \text{ m})^2 = 0.0707 \text{ m}^2$$

$$a) \quad P - P_0 = \frac{W}{A} = \frac{45 \text{ N}}{0.0707 \text{ m}^2} = \underline{636 \text{ Pa}}$$

$$b) \quad i) \Delta P = \frac{83 \text{ N}}{0.0702 \text{ m}^2} = \underline{1170 \text{ Pa}} \quad ii) \underline{1170 \text{ Pa}}$$

12.2) Elevador hidráulico I

$$A = \frac{\pi D^2}{4} \quad mg = \frac{\pi D_2^2/4}{\pi D_1^2/4} F_1 \cdot \frac{D_2}{D_1} = \sqrt{\frac{mg}{F_1}} = \sqrt{\frac{(1520 \text{ kg})(9.8 \text{ m/s}^2)}{725 \text{ N}}} = 0.9$$

Sección 12.3 Rotación

12.25) $V = \frac{m}{\rho}$ $dA = V$

$$V = \frac{m}{\rho} = \frac{70 \text{ kg}}{1030 \text{ kg/m}^3} = 0.06796 \text{ m}^3 \quad d = \frac{V}{A} = \frac{0.06796}{\pi(0.45)^2} =$$

$$d = 0.107 \text{ m}$$

12.27) $W = mg = 17.5 \text{ N}$ $m = 1.79 \text{ kg}$

$$T + B - mg = 0 \quad B = mg - T = 17.5 \text{ N} - 11.20 \text{ N} = 6.3 \text{ N}$$

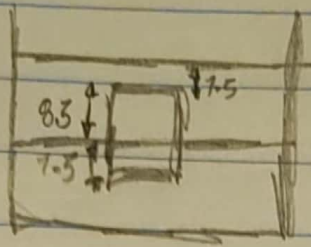
$$V_{\text{objeto}} = \frac{B}{\rho g} = \frac{6.3}{(1 \times 10^3)(9.8)} = 6.43 \times 10^{-4} \text{ m}^3$$

$$\rho = \frac{m}{V} = \frac{1.79 \text{ kg}}{6.43 \times 10^{-4} \text{ m}^3} = 2.78 \times 10^3 \text{ kg/m}^3$$

12.29) a) El que se desplaza tiene menor volumen que el objeto, pero deben pesar lo mismo.

b) Porque en promedio la densidad del barco es menor a la del agua.

12.31) a)



$$P - P_0 = \rho g h$$

$$P - P_0 = (790 \text{ kg/m}^3)(9.8 \text{ m/s}^2)(0.015 \text{ m})$$

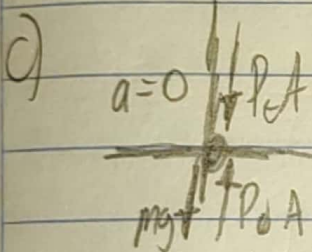
$$P - P_0 = 116 \text{ Pa}$$

$$b) P = P_a + P g (0.100 \text{ m})$$

$$P - P_a = P g (0.100 \text{ m}) + P g (0.015 \text{ m})$$

$$P - P_a = [(790 \text{ kg/m}^3)(0.100 \text{ m}) + (1000 \text{ kg/m}^3)(0.015 \text{ m})](9.8 \text{ m/s}^2)$$

$$P - P_a = 921 \text{ Pa}$$



$$\sum F_y = m a_y$$

$$P_0 A - P_b A - mg = 0$$

$$(P_0 - P_b) A = mg$$

$$m = \frac{(P_0 - P_b) A}{g} = \frac{(805 \text{ Pa})(0.0100 \text{ m}^2)}{9.8 \text{ m/s}^2} = 0.821 \text{ kg}$$

$$\rho = \frac{m}{V} = 0.821 \text{ kg} / (0.100 \text{ m})^3 = 821 \text{ kg/m}^3$$

$$12.33) \frac{7.102 \text{ kg}}{1 \times 10^{-3} \text{ kg/m}^3} = 7.102 \times 10^{-3} \text{ m}^3$$

$$39.2 \text{ N} - 78.6 \text{ N} = -20.6 \text{ N} \quad 20.6 \text{ N} / 9.8 \text{ m/s}^2 = 2.102 \text{ kg}$$

$$2.102 \text{ kg} / (7.102 \times 10^{-3} \text{ m}^3) = 2.91 \times 10^3 \text{ kg/m}^3$$

Sección 12.4 Flujo de fluido

$$12.35) V_2 = V_1 \left(\frac{A_1}{A_2} \right)$$

$$A_1 = \pi (0.8 \text{ m})^2$$

$$A_2 = 20\pi (0.7 \text{ m})^2$$

$$V_2 = 3 \text{ m/s} \cdot \frac{\pi (0.8)^2}{20\pi (0.7)^2} = 9.6 \text{ m/s}$$